

 **Jim Cramer** @jimcramer · Jan 29

Don't you love a stock that goes down when a co-founder ENDORSES its ceo? The same thing happened at **\$MCD** before the firing.
\$TWTR

🔄 12 ★ 24 ⋮

#CASHTAG

BIG DATA PIPELINE FOR USER SENTIMENT ANALYSIS

Shafi Bashar



 **CNBC** @CNBC · Jan 30

Apple just hit a new high and is now worth \$705,816,360,000. Why **\$AAPL** will go even higher: cnb.cx/1BqTK3T

MOTIVATION

- People have opinions
- Different sources, different mediums - Twitter, Reddit, Facebook etc.
- Platform for aggregating opinions and analyzing on a Topic
- v 1.0: User's opinion of US stock market

DEMO

- Webpage

<http://www.hashtagcashtag.com>

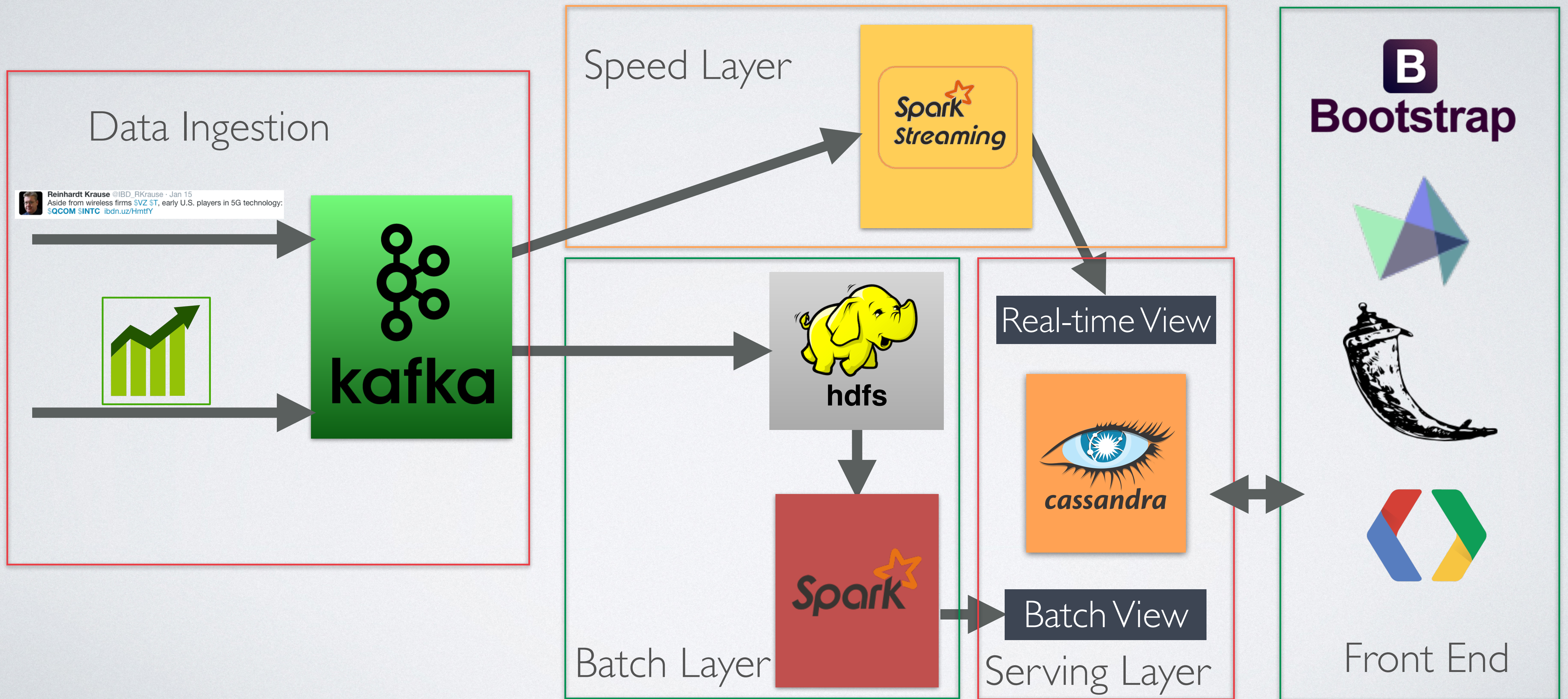
- Video

<http://youtu.be/7oMrj7nIHr4>

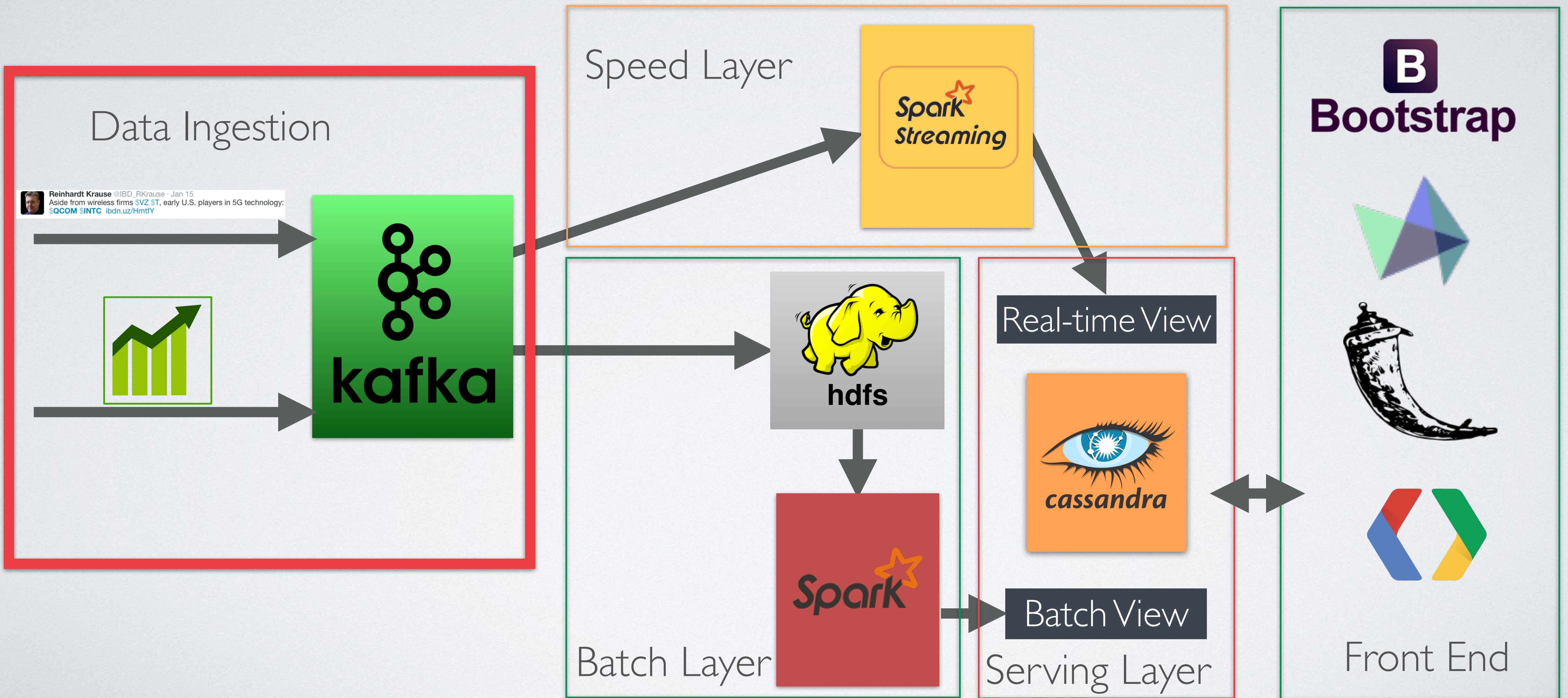
- Alternate Link

<http://54.67.108.50>

PIPELINE λ ARCHCHITECTURE



PIPELINE λ ARCHCHITECTURE



DATA INGESTION

- Twitter Data from streaming API
- Limited to 250 stocks

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DATA INGESTION

- Stock Data from www.netfonds.no
- delayed by 1 hour
- need to fetch for each individual stocks
- In Oslo time
- Incremental CSV file

20150126T153000	113.62	297	Auto trade
20150126T153000	113.66	375	Auto trade
20150126T153000	113.67	100	Auto trade
20150126T153000	113.65	161	Auto trade
20150126T153000	113.68	270	Auto trade
20150126T153000	113.67	100	Auto trade
20150126T153001	113.66	100	Auto trade
20150126T153001	113.65	100	Auto trade
20150126T153001	113.67	100	Auto trade
20150126T153001	113.65	100	Auto trade
20150126T153001	113.66	100	Auto trade
20150126T153001	113.67	100	Auto trade
20150126T153001	113.68	254	Auto trade
20150126T153001	113.65	100	Auto trade
20150126T153001	113.67	100	Auto trade
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20150126T153001	113.67	100	Auto trade

DATA INGESTION

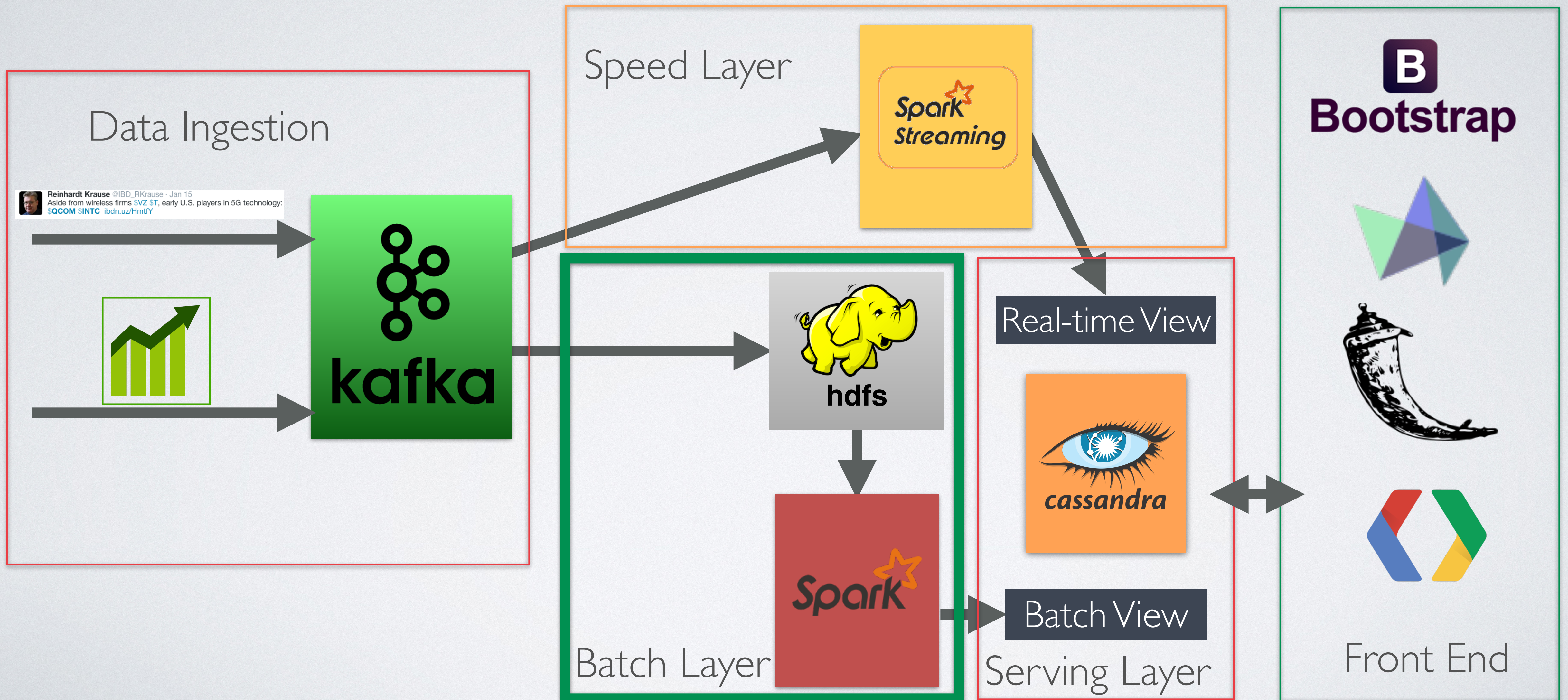
- Stock Data from www.netfonds.no
- Pre-processing

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DATA INGESTION

- Multi topic, multi consumer Kafka
- Topic - Twitter and Stock
- Consumers - HDFS and Streaming for each topic

PIPELINE λ ARCHCHITECTURE



BATCH LAYER

- Spark batch job (codes in scala)
- Twitter
 - Number of mentions/time granularity
 - Sentiment of the mention
 - Top trending stocks

ticker	year	month	day	hour	minute	frequency	sentiment
TSLA	2015	2	3	16	35	16	0
TSLA	2015	2	3	16	34	1	0
TSLA	2015	2	3	16	33	1	0
TSLA	2015	2	3	16	32	1	0
TSLA	2015	2	3	16	31	9	3
TSLA	2015	2	3	16	30	4	0
TSLA	2015	2	3	16	28	2	2
TSLA	2015	2	3	16	27	4	0
TSLA	2015	2	3	16	25	4	0
TSLA	2015	2	3	16	24	4	0
TSLA	2015	2	3	16	23	9	0
TSLA	2015	2	3	16	20	4	0
TSLA	2015	2	3	16	19	36	0
TSLA	2015	2	3	16	16	4	0
TSLA	2015	2	3	16	14	1	0
TSLA	2015	2	3	16	6	1	1
TSLA	2015	2	3	16	2	1	1
TSLA	2015	2	3	16	1	1	0
TSLA	2015	2	3	16	0	9	0
TSLA	2015	2	3	15	58	1	-1
TSLA	2015	2	3	15	57	4	4
TSLA	2015	2	3	15	56	1	0
TSLA	2015	2	3	15	55	1	2
TSLA	2015	2	3	15	50	9	0

SENTIMENT ANALYSIS

Positive

"upgrade",
"upgraded",
"long",
"buy",
"buying",
"growth",
"good",
"gained",
"well",
"great",
"nice",
"top",
...

Negative

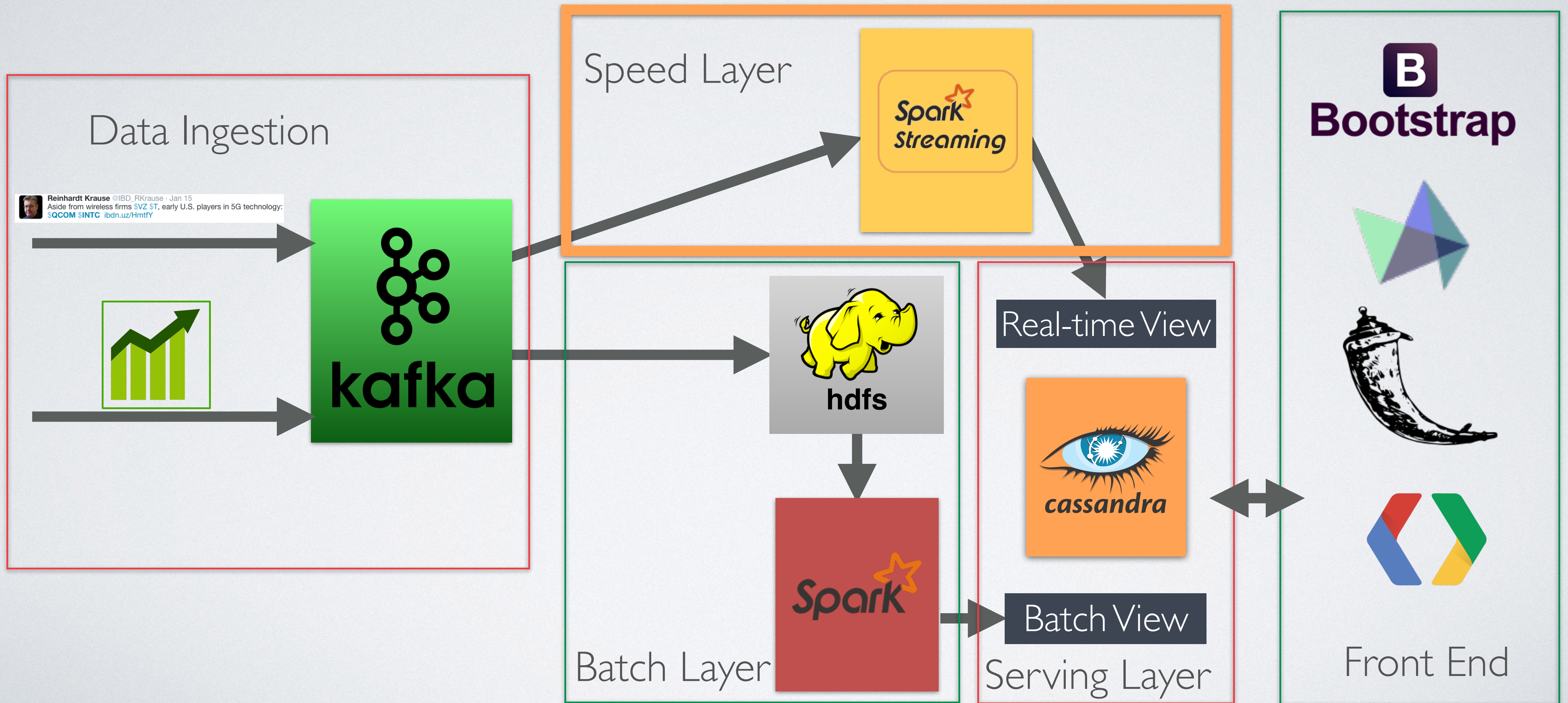
"downgraded",
"bears",
"bear",
"bearish",
"volatile",
"short",
"sell",
"selling",
"forget",
"down",
"resistance",
"sold",
...

BATCH LAYER

- Azkaban controls the flow and scheduling
- Re-computation Algorithm
- Stocks
 - high, low, open, close, volume
 - different granularity

ticker	year	month	day	hour	minute	close	high	low	open	volume
TWTR	2015	1	30	13	0	37.55	37.55	37.55	37.55	6740
TWTR	2015	1	30	12	59	37.54	37.56	37.46	37.47	96070
TWTR	2015	1	30	12	58	37.47	37.51	37.46	37.51	47839
TWTR	2015	1	30	12	57	37.5	37.55	37.495	37.54	65830
TWTR	2015	1	30	12	56	37.54	37.57	37.54	37.565	41758
TWTR	2015	1	30	12	55	37.565	37.6	37.5	37.5	54317
TWTR	2015	1	30	12	54	37.5	37.54	37.5	37.52	36129
TWTR	2015	1	30	12	53	37.525	37.56	37.5	37.509	18245
TWTR	2015	1	30	12	52	37.5	37.52	37.49	37.5	25879
TWTR	2015	1	30	12	51	37.505	37.51	37.48	37.485	35091
TWTR	2015	1	30	12	50	37.49	37.523	37.475	37.523	29660
TWTR	2015	1	30	12	49	37.52	37.57	37.52	37.55	15814
TWTR	2015	1	30	12	48	37.55	37.59	37.55	37.56	10457
TWTR	2015	1	30	12	47	37.55	37.6	37.53	37.586	19380
TWTR	2015	1	30	12	46	37.58	37.6	37.55	37.59	21223
TWTR	2015	1	30	12	45	37.59	37.62	37.54	37.55	26665
TWTR	2015	1	30	12	44	37.54	37.56	37.51	37.52	15934
TWTR	2015	1	30	12	43	37.52	37.56	37.48	37.55	49035
TWTR	2015	1	30	12	42	37.55	37.58	37.55	37.56	29777
TWTR	2015	1	30	12	41	37.55	37.63	37.55	37.63	36928
TWTR	2015	1	30	12	40	37.63	37.67	37.62	37.67	13526
TWTR	2015	1	30	12	39	37.67	37.69	37.645	37.69	10726
TWTR	2015	1	30	12	38	37.695	37.745	37.68	37.726	25958
TWTR	2015	1	30	12	37	37.72	37.73	37.7	37.72	8619

PIPELINE λ ARCHCHITECTURE



SPEED LAYER

- Incremental Algorithm (codes in Scala)
- Number of mentions/sentiments at different granularity
 - Similar operation as in batch layer
- Use the reduce operation 'updateStateByKey'

Batch Operation

Batch Operation

Speed|Speed|Speed|Speed|Speed

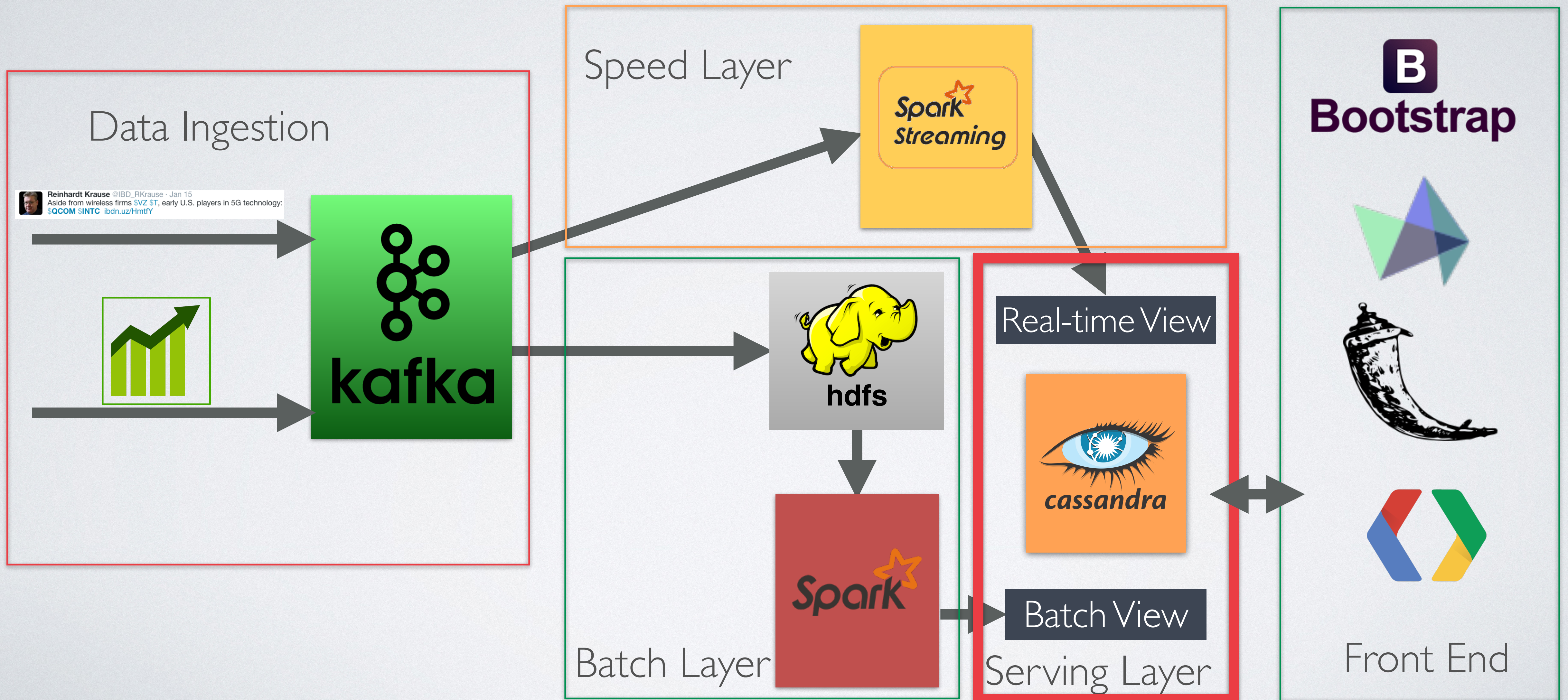
data over time

SPEED LAYER

- Rolling Count Operation: Top trending in the last 10 minutes updated every 5 seconds
- Use the reduce operation 'reduceByKeyAndWindow'

id	timestamp	frequency	sentiment	ticker
0	1430375561	55	-5	AAPL
0	1430370589	55	-5	AAPL
0	1430365508	54	-5	AAPL
0	1430360540	54	-5	AAPL
0	1430355643	53	-5	AAPL
0	1430350571	53	-5	AAPL
0	1430345558	52	-5	AAPL
0	1430340632	52	-7	AAPL
0	1430335545	52	-7	AAPL
0	1430330620	52	-7	AAPL
0	1430325548	55	-9	AAPL
0	1430320555	55	-9	AAPL
0	1430315608	55	-9	AAPL
0	1430310541	55	-9	AAPL
0	1430305549	55	-9	AAPL
0	1430300622	55	-9	AAPL
0	1430295556	55	-9	AAPL

PIPELINE λ ARCHCHITECTURE



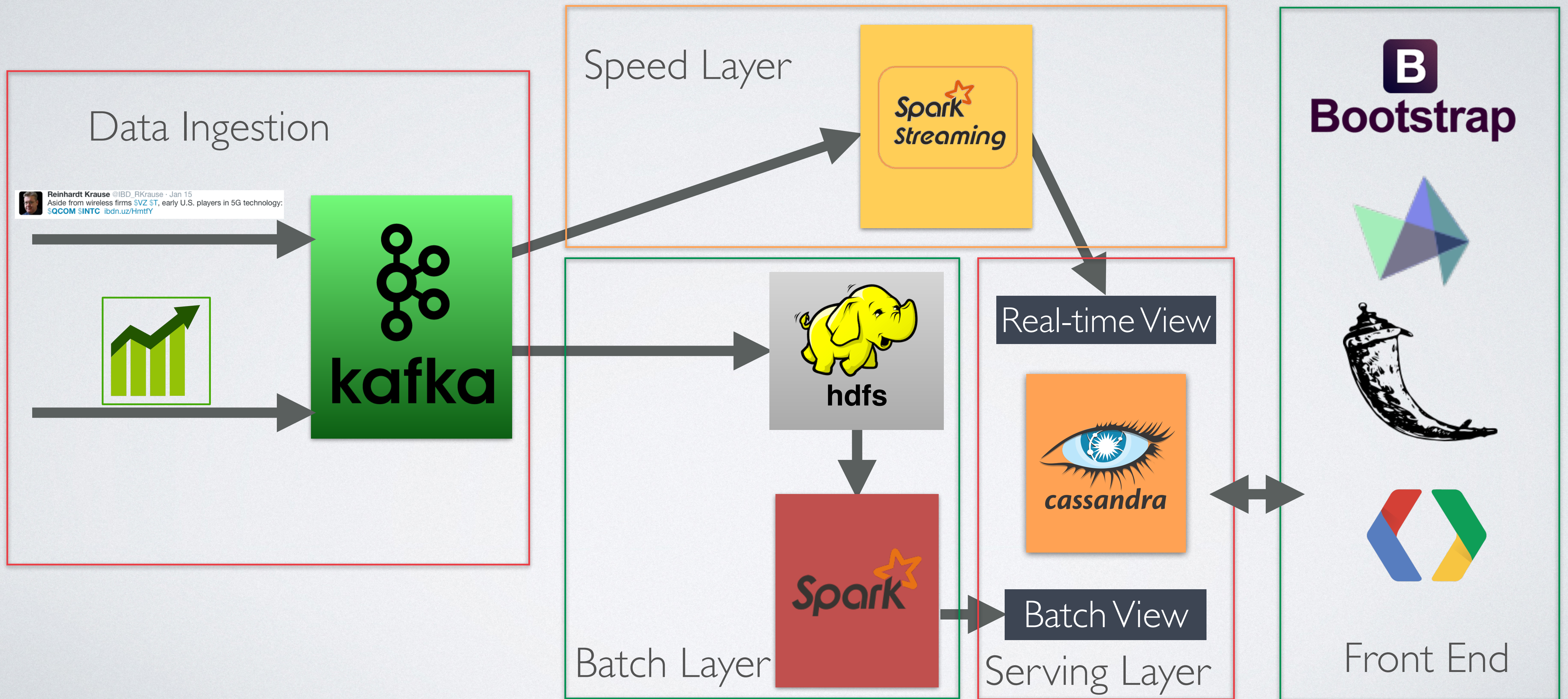
SERVING LAYER

- De-normalized tables in Cassandra
- Twitter Time Series
 - partitioned by ticker symbol
 - clustering order by (year, month, day, hour, minute)
- Top Trending Stocks
 - partitioned by (year, month, day, hour)
 - clustering order by number of mentions

SPEED VIEW

- Cassandra TTL support can be used for rolling count operation for dashboard application
- Not available in Cassandra-Spark connector
- Add timestamp and ranking to each ticker generation in each 5 second window
- Partitioned by ranking, clustering order by timestamp

PIPELINE λ ARCHCHITECTURE



SHAFI BASHAR



- PhD, ECE, UC Davis
- Present - Intel Corporation
- Worked on 4G LTE, WiFi standardization
- Interest -Algorithm, Machine Learning
- Activities - backpacking, skiing, running, photography

